

CSE423 Assignment 4

- ① a) HSV represents the Hue, saturation and value of a specific color and is geometrically shown as a cone where the middle top value is shown as white. HLS model represents the hue, saturation and the perceived brightness L of a color and is shown as a double cone where the middle of the cone is gray and gets white as it goes up. HSV is preferred when user want to adjust the hue without affecting the brightness of the color (V) overall. HSL is preferred when the perceived brightness needs to be altered without affecting the color character overall, HSL is more friendly towards changing of the tint / shade independently.

(b) Given, $H = 45^\circ$, $S = 0.8$, $V = 0.9$

$$\text{now, } C = V \times S = 0.72$$

$$X = C \times (1 - |(H/60 \% 2) - 1|)$$

$$= 0.72 \times (1 - |(45/60 \% 2) - 1|)$$

$$= 0.72 \times \frac{3}{4} = 0.54$$

$$m = V - C$$

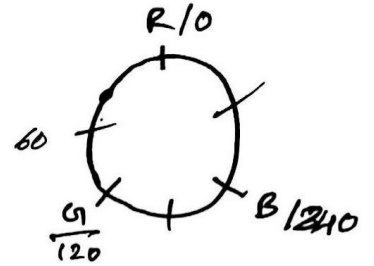
$$= 0.18$$

$$\text{As } H = 45^\circ, \quad R, G, B = C, X, 0 + [m]$$

$$= 0.72, 0.54, 0$$

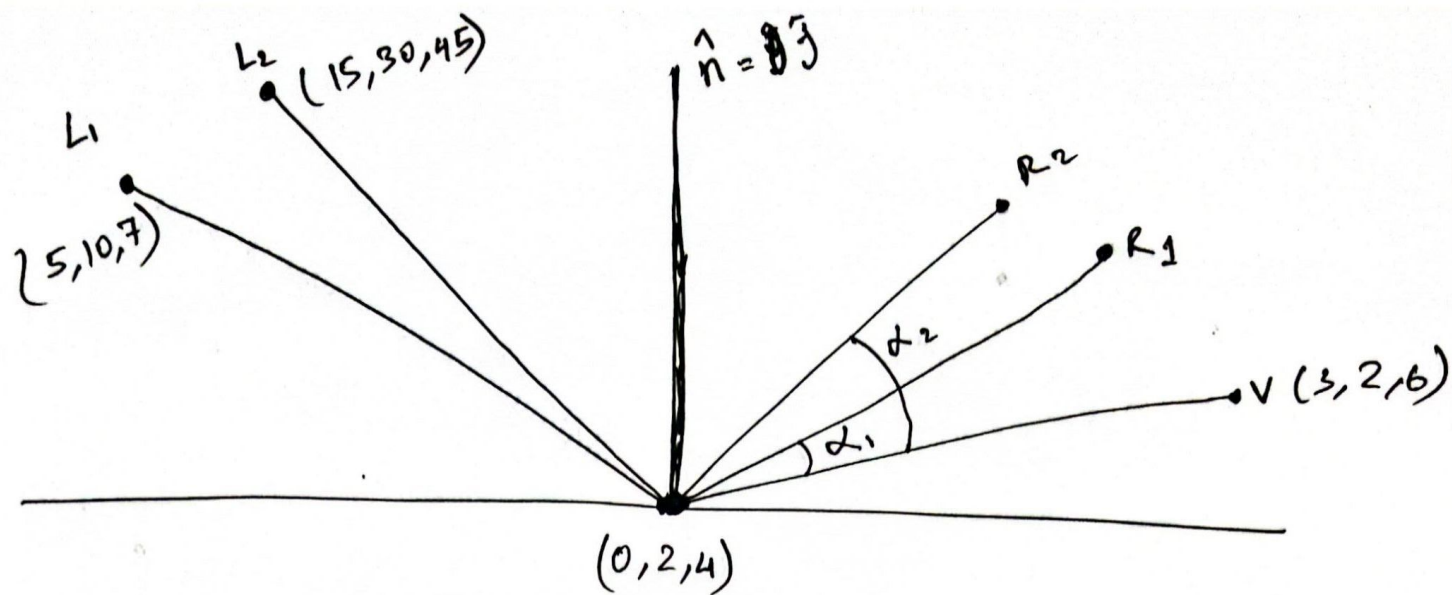
$$= 0.9, 0.72, 0.18$$

$$\therefore CMY = 1 - RGB = 0.1, 0.28, 0.82$$



②

(a)



Ambient Intensity = $8 \times 0.2 = 1.6$

$$\vec{V} = (3, 2, 6) - (0, 2, 4) = (3, 0, 2)$$

$$\hat{V} = \frac{3i + 2k}{\sqrt{13}} = \frac{1}{\sqrt{13}} (3i + 2k)$$

Light - 1 (L_1):

$$\vec{L}_1 = (5, 10, 7) - (0, 2, 4) = (5, 8, 3)$$

$$\hat{L}_1 = \frac{5i + 8j + 3k}{\sqrt{98}}$$

$$\hat{L}_1 \cdot \hat{n} = \hat{L}_1 \cdot \hat{j} = \frac{8}{\sqrt{98}}$$

$$\vec{R}_1 = 2(\hat{L}_1 \cdot \hat{n})\hat{n} - \vec{L}_1 = 2 \times \frac{8}{\sqrt{98}} \hat{j} - \vec{L}_1$$

$$= -\frac{5}{\sqrt{98}} i - \frac{3}{\sqrt{98}} k + \left(\frac{16}{\sqrt{98}} - \frac{8}{\sqrt{98}} \right) j$$

$$= -\frac{5}{\sqrt{98}} i + \frac{8}{\sqrt{98}} j - \frac{3}{\sqrt{98}} k$$

$$R_1 \cdot \hat{V} = \frac{1}{\sqrt{98}} (-5i + 8j - 3k) \times \frac{1}{\sqrt{13}} (3i + 2k) = -0.588$$

= -ve = 0

$$\begin{aligned} \therefore d_1 &= \sqrt{98} \\ \therefore a_1 &= 1 - \left(\frac{d_1}{r} \right)^2 \\ &= 1 - \left(\frac{\sqrt{98}}{25} \right)^2 \\ &= 0.8432 \end{aligned}$$

~~$\therefore$ Diffuse + specular from light 1 =~~

Light - 2 (L_2):

$$\vec{L_2} = (15, 30, 45) - (0, 2, 4) = (15, 28, 41)$$

$$\hat{L_2} = \frac{15\mathbf{i} + 28\mathbf{j} + 41\mathbf{k}}{51.9}$$

$$\hat{L_2} \cdot \hat{n} = 0.539$$

$$\therefore \vec{R_2} = 2 \cdot (0.539 \mathbf{j}) - \left(\frac{15\mathbf{i} + 28\mathbf{j} + 41\mathbf{k}}{51.9} \right)$$

$$= -0.289\mathbf{i} + 0.539\mathbf{j} - 0.790\mathbf{k}$$

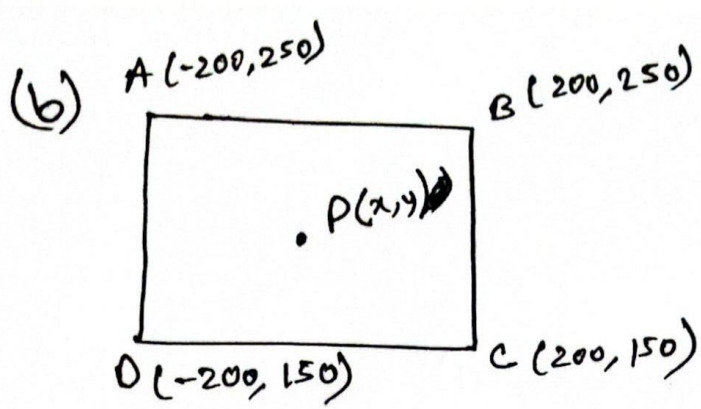
$$R_2 \cdot \vec{V} = \left(\frac{3}{\sqrt{13}}\mathbf{i} + 0\mathbf{j} + \frac{2}{\sqrt{13}}\mathbf{k} \right) \cdot R_2$$

$$= -ve = 0$$

$$\therefore d_2 = \sqrt{15^2 + 28^2 + 41^2} = 51.9$$

$$\therefore a_2 = 1 - \left(\frac{51.9}{60} \right)^2 = 0.253$$

$$\begin{aligned} \therefore \text{Final intensity} &= 1.6 + 20 \times 0.843 \left(0.6 \times \frac{8}{\sqrt{98}} + 0.4 \times (0)^{10} \right) \\ &\quad + 10 \times 0.253 \left(0.6 \times 0.539 + 0.4 \times (0)^{10} \right) \\ &= 10.60 \end{aligned}$$



$$A_{RAB} = (0.7, 0.2, 0.5)$$

$$C_{RAB} = (0.8, 0.7, 0.6)$$

P is the center of A and C

$$\therefore P_x = 0$$

$$P_y = \frac{150 + 250}{2} = 200$$

Now, as P is equidistant from A and C ,

$$P_R = \frac{0.7 + 0.8}{2} = 0.75$$

$$P_A = \frac{0.2 + 0.7}{2} = 0.45$$

$$P_B = \frac{0.5 + 0.6}{2} = 0.55$$